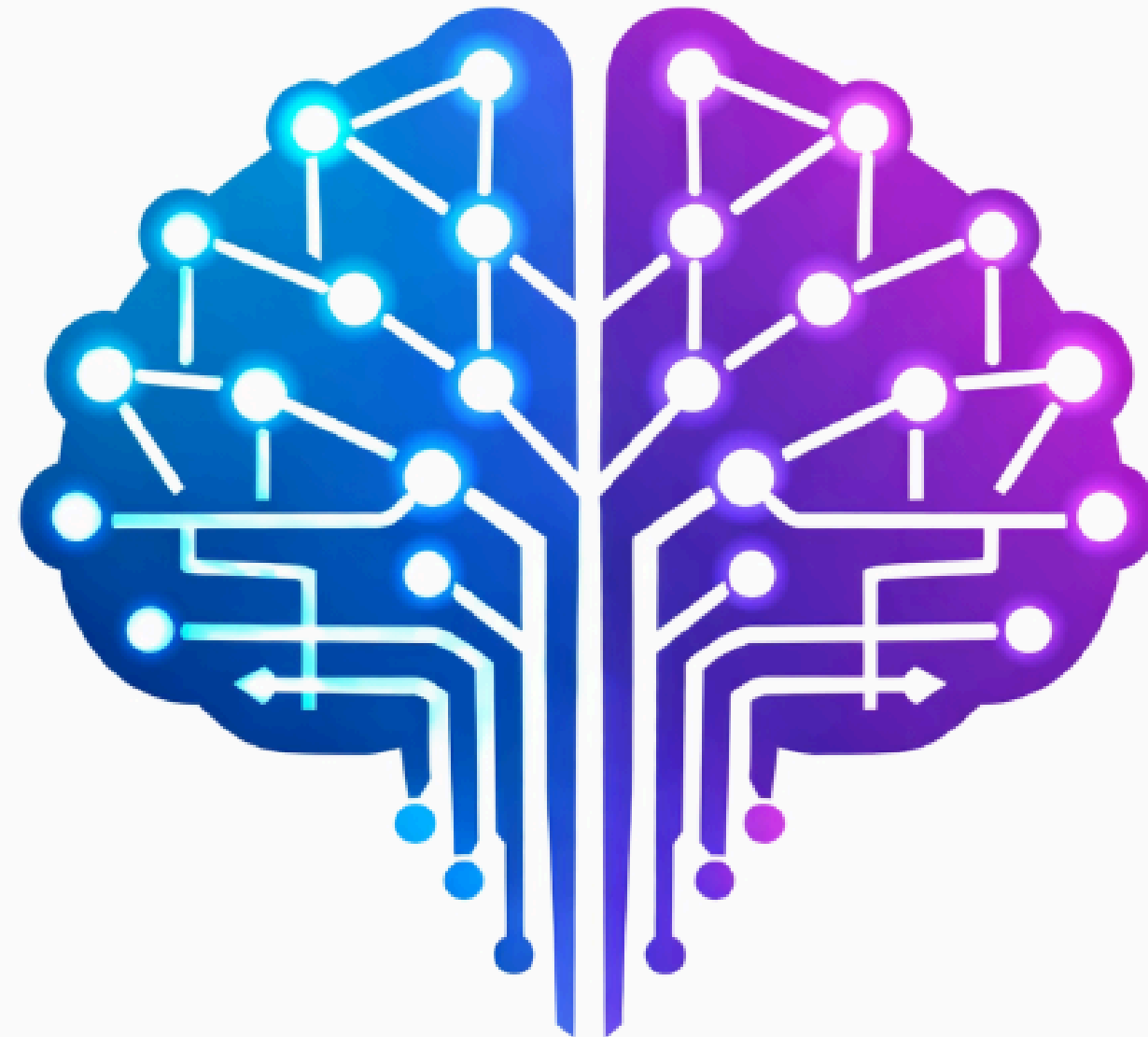


GENAI FUNDAMENTALS



LECTURE:

APIs **VS LOCAL MODELS (REAL-WORLD DECISIONS)**

Two Ways to Use LLMs

👉 Developers typically interact with LLMs in **two ways**:

API-based models

- Accessed through cloud APIs

Local / Self-hosted models

- Run on your own infrastructure

Both approaches are widely used in real-world applications.

How API-Based LLMs Work

☞ API-based models are hosted by providers.

Workflow:

Application → API Request → Model → Generated Response

Benefits:

- easy integration
- no infrastructure management
- access to powerful models

Limitations of API-Based Models

API models also come with tradeoffs:

- usage cost
- dependency on external providers
- limited customization
- data privacy concerns

Developers must consider these factors in production systems.

How Local LLMs Work

Local models run directly on your own infrastructure.

Typical setup:

Application → Local Model Server → Generated Response

Examples of local model tools:

- Ollama
- vLLM
- Hugging Face Transformers
- LM Studio

Advantages of Local Models

Local models provide several benefits:

- full data privacy
- no per-request API costs
- greater customization
- offline capability

👉 This is important for many enterprise use cases.

Challenges of Running Local Models

Local models introduce new challenges:

- GPU requirements
- infrastructure management
- model optimization
- scaling complexity

👉 Operating LLM infrastructure can be expensive and complex.

How Developers Make the Decision

Choosing between APIs and local models depends on:

- privacy requirements
- budget constraints
- infrastructure availability
- performance needs
- customization goals

Many real-world systems use a **hybrid approach**.

